

Project 4: Preference Elicitation

Task 1: Movie Feature Representation

For this project, I used the IMDB movie dataset. I represented each movie using its **IMDb score, release year and genres**.

The IMDb score and release year were scaled between 0 and 1 using MinMaxScaler. Since one movie can have more than one genre, I separated the genres and converted them into binary columns using one-hot encoding.

I also kept the movie title so that it can be shown in the interface, but the title is not used as a numerical feature.

I chose these features because they give useful information about a movie while keeping the representation simple.

Task 2: Preference Model

The Bradley-Terry model is normally used to compare two items.

For a movie i , the utility is:

$$U(i) = w^T x_i$$

where x_i is the movie feature vector and w represents the user's preferences.

For two movies i and j , the probability that movie i is preferred is:

$$P(i > j) = \exp(U(i)) / (\exp(U(i)) + \exp(U(j)))$$

For Design 2, the user ranks several movies instead of comparing only two. For example:

$$i_1 > i_2 > i_3 > \dots > i_n$$

I use a sequential extension of the model:

$$P(i_1 > i_2 > \dots > i_n) = \prod_{k=1}^{n-1} \frac{\exp(U(i_k))}{\sum_{j=k}^n \exp(U(i_j))}$$

The idea is simple. First, the model considers the first movie as the preferred movie from the full set. Then the second movie is preferred from the remaining movies, and this continues until the ranking is complete.

This makes it possible to use the same preference idea for both pairwise choices and rankings.

Task 3: User Study

The goal of the user study is to compare two different ways of collecting movie preferences.

Design 1: Pairwise Comparison

In Design 1, the user is shown two movies and chooses the one they would rather watch.

The interface shows the movie title, year, IMDb score and genres to help the user make the choice.

Design 2: Top-10 Ranking

In Design 2, the user is shown ten movies.

The user can drag and reorder the movies from:

1 = most preferred

to:

10 = least preferred

The final ranking is then submitted.

Hypothesis

My hypothesis is that **Design 1 will be easier to use** because the user only has to compare two movies at a time.

However, **Design 2 may collect more preference information in fewer interactions** because the user ranks ten movies at once.

Participants and Procedure

If I carried out the study, I would recruit students or other volunteers who are familiar with movies.

Each participant would try both designs. To make the comparison fair, some participants would use Design 1 first and others would use Design 2 first.

After completing both interfaces, I would ask them about their experience.

I would compare:

- time needed to complete the task
- difficulty of each interface
- number of interactions
- which design the participant preferred

Conclusion

In this project, I created two ways of collecting movie preferences.

Design 1 uses pairwise comparisons, while Design 2 allows the user to rank ten movies.

The movie features are based on IMDb score, release year and genres. The proposed user study can then be used to compare which interface is easier and which one collects preference information more efficiently.